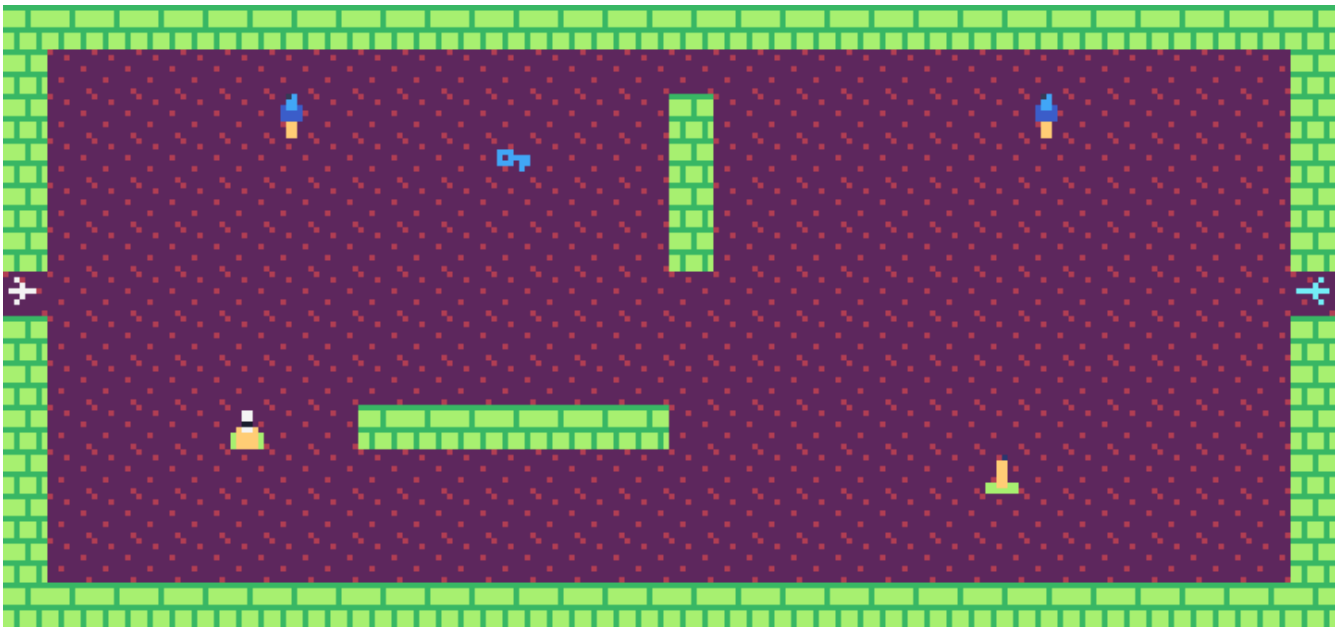
STUDENT HANDBOOK / TIC-80 + PYTHON

ESCAPE THE CRYPT

Build your own retro adventure, one Python condition at a time.

YOUR MISSION

Explore three rooms, speak to characters, collect a key, activate a lever and unlock the exit. The starter game runs, but its puzzles are unfinished. You will complete them.



MORNING	AFTERNOON
Explore the editor, redraw the hero and customise rooms	Learn variables and conditions; program the key, lever and exit
Add new characters and conversations	Playtest your game with another student

No Lua, no external game framework. The game uses the Python runtime built into recent TIC-80 versions.

01 Start the game

Open **https://tic80.com/create** (or a recent TIC-80 desktop build). You will edit the **.tic cartridge**, not run the Python source with your computer's normal Python interpreter.

Import the starter cartridge

At the TIC-80 console, enter:

```text
add
dir
load Escape_the_Crypt_Starter_Python
run
```

The **add** command opens a file picker in the browser version. Select **Escape_the_Crypt_Starter_Python.tic** from your student kit. If the cart is already in the TIC-80 directory, skip **add**.

Controls and editor shortcuts

Shortcut	Action
Arrow keys	Move the hero
Z	Talk to nearby people / use an object
F1	Edit Python code
F2	Draw sprites (characters and tiles)
F3	Edit the map (three rooms)
Ctrl + R	Run the current game
Ctrl + S	Save your changes
Ctrl + F (inside CODE)	Search for MISSION

Save your own copy

Open the console with **Esc**, then use a unique cartridge name before you begin changing the game:

```text
save My_Crypt.tic
```

Browser version: use **get My_Crypt.tic** in the console to download your saved cartridge.

If you still see HELLO WORLD, the starter cartridge was not loaded. Run the load command above before run.

02 Design your crypt

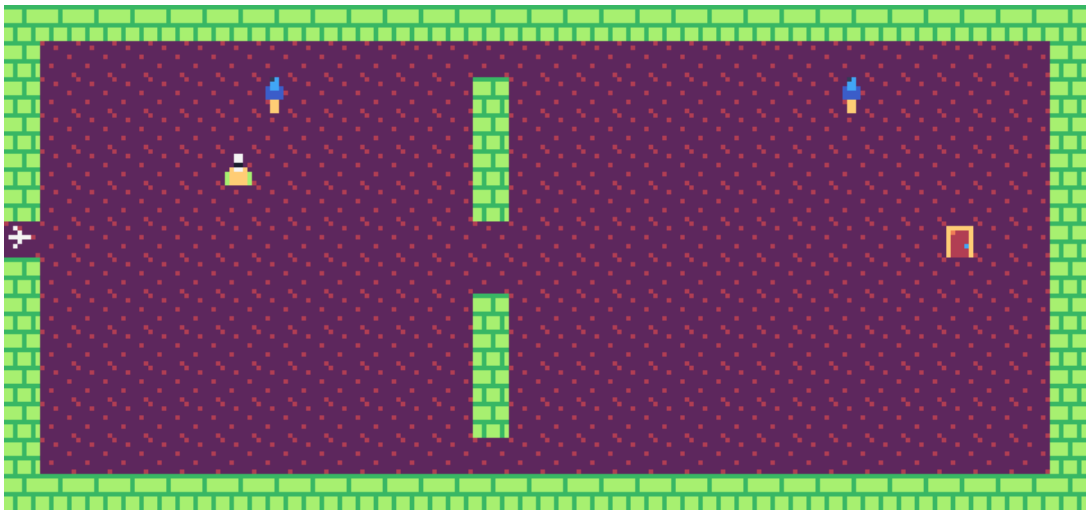
Mission 1 - Create your own hero

Open **SPRITES (F2)**. Select **sprite #256** (the first foreground sprite), then draw your character. It can be a knight, a skeleton, a detective or anything you invent. The code already uses this sprite.

```python
HERO = 256
# In draw_game():
spr(HERO, player_x * 8, player_y * 8, 0)
```

Mission 2 - Edit the three rooms

Open **MAP (F3)**. Three complete rooms already exist side by side. You can rearrange the furniture, walls and objects. Each room is **30 tiles wide and 14 tiles high**. The room areas begin at columns 0, 32 and 64 of the large map.



Tile ID	What it does
0 - FLOOR	Walkable ground
1 - WALL	Blocks movement
3 - NPC	Character you can speak to
4 - KEY	Collectible key
5 - LEVER	Interactive mechanism
6 - DOOR	Final exit
7 / 8 - EXITS	Travel between rooms

Important: walls, objects and characters stop the player. Keep at least one walkable route between the objects and the room exits.

03 Learn the Python basics

Press **F1** to open CODE, then **Ctrl + F** and search for **MISSION**. Read the comments marked with **#**: they are the exercises you will complete.

Variables remember facts

```
```python
has_key = False
lever_on = False

# Later, when the player collects the key:
has_key = True
```
```

True and **False** are Boolean values. The status bar reads these variables and displays the current game state.

Conditions make choices

```
```python
if has_key:
    show_dialog(["You found the key!"])
else:
    show_dialog(["Find the key first."])
```
```

In Python, the colon **:** starts the block and **indentation matters**. Use four spaces for statements inside **if** and **else**.

Mission 3 - Write better dialogue

Find the **if tile == NPC** part inside **interact()**. Replace the existing lines with your own story. Add a conditional line that changes after the key is collected.

```
```python
if has_key:
    show_dialog(["You are getting closer!"])
else:
    show_dialog(["Look for something useful."])
```
```

Try changing the room names in ROOMS near the top of the file. Your new names will appear in the status bar.

04 Make the world interactive

Mission 4 - Pick up the key

Search for **MISSION 4** in **interact()**. When the hero stands next to the key and presses Z, the game should:

- Remember that the player has the key.
- Remove the key tile from the room.
- Show a short pickup message.

TIC-80 gives you two functions for the map: **mget(x, y)** reads a tile ID; **mset(x, y, tile_id)** replaces it. The local tile coordinates must be offset by the current room's map position.

```
```python
# A map-edit example (not the key solution)
map_x = ROOMS[room_index][1]
mset(map_x + x, y, FLOOR)
```
```

Mission 5 - Pull the lever

Search for **MISSION 5**. Make the lever react to Z. Use the **lever_on** variable to remember that it has been activated, and replace its tile with **LEVER_ON**.

```
```python
# Example of changing a Boolean variable
lever_on = True
```
```

CHECKPOINT

Can you pick up the key, see KEY:YES at the bottom of the screen, pull the lever and see LEVER:ON? If so, you are ready for the final puzzle.

05 Unlock the exit

Mission 6 - Win only with the key AND the lever

Find **MISSION 6**. The door must stay closed until **both** conditions are satisfied. Otherwise, show a message explaining what is missing.

```
```python
if has_key and lever_on:
    # Set the victory variable to True here
    pass
elif not has_key:
    # Show a missing-key hint here
    pass
else:
    # Show a missing-lever hint here
    pass
```
```

The word pass is a placeholder. Replace it with your own code.

Optional challenges

- Create a fourth room and a new character.
- Add a secret ending, chosen through a different condition.
- Write a riddle: a character gives a hint only after you collect an item.
- Change sprite colours, room layouts and dialogue to give your game its own atmosphere.

Final playtest

- All three rooms are reachable. No object blocks an essential route.
- All characters respond when you press Z next to them.
- The key disappears when collected; the status bar updates.
- The lever changes visually and the status bar updates.
- The door refuses to open without the prerequisites, then allows a win.
- Another student can play your game without an explanation.

Submit your game

```
```text
save My_Crypt.tic
get My_Crypt.tic
```
```

References: https://tic80.com/create | https://dev.tic80.com/learn | https://github.com/nesbox/TIC-80/wiki/supported-languages